

AB Pharma

Pharmaceutical company A acquired another pharmaceutical company B and the resultant company needed a tool for managing the data/documents. The data was present as documents and the primary use of the tool is to upload these documents and transfer it to regulatory committee. There was an existing tool used by company B; this was basically a java applet to simply upload a package (which has to be zipped) before uploading which did not help the parent company as their standard are different. So altogether a new tool was required.

Assessment

USER INTERVIEW:

Employee from Company A

- Q) How often do you need to upload the documents.
- A) We have documents from each medical area to be send for standardisation to the regulatory committee. This can be in range of 5-20 different set of documents to be send every day from each area.
- Q) What would be the package size range
- A) This can be in the range of 1 GB – 80 GB per directory
- Q) Does all the package have same extension? If not, why?
- A) No, the package extension varies due to locality of the regulatory committee. We have pharma products from different parts of the globe, so each country/region will have a different standard.
- Q) Could you confirm on the various different regulatory committees
- A) We have Europe, Central America, Latin America and APAC regions.
- Q) Can you confirm if a single user uploads/monitor for multiple regions?
- A) No, each user is assigned to a distinct region.
- Q) What other feature would you like to add, other than the primary feature (i.e. upload)
- A) Preferably tracking the status of the package.
- Q) Would you like to have an SSO login (so that the LDAP credentials can be used)
- A) Yes, that would eventually help.

Employee from Company B

Q) How often do you need to upload the documents.

A) We have to upload 5-10 zip files every day.

Q) What would be the package size range

A) This can be in the range of 1-5GB

Q) Does all the package have same extension? If not, why?

A) Yes, we only compress the directory to a zip file and upload to the Central American regulatory.

Q) Could you confirm on the various different regulatory committees

A) NA

Q) Can you confirm if a single user uploads/monitor for multiple regions?

A) NA

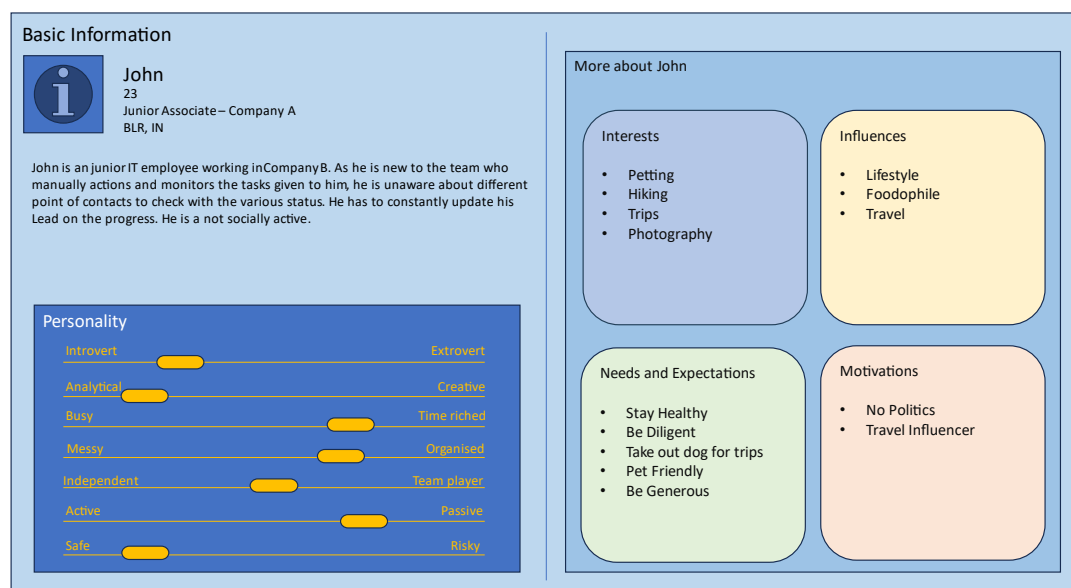
Q) What other feature would you like to add, other than the primary feature (i.e. upload)

A) We would like to have the folder to be automatically zipped and uploaded since we do this manually now.

Q) Would you like to have an SSO login (so that the LDAP credentials can be used)

A) Yes

User Persona:



Cognitive Research

We performed the User experience Interviews to arrive at the below conclusions:

Requirements:

We have come up with the below requirements to be included in creating the tool:

- Tool to automatically create the package and send it to respective regulatory committee (Depending on the region).
- Tool to track the progress of packaging/uploading.

Pain Points:

After Interviewing few of the employees from both the companies, we have learnt the below pain points of their day-to-day tasks.

- Employees find it hard to manually zip (create the package) and send it across to the regulatory.
- It is difficult to track the status of the package after sending it as it involves communication with multiple teams and vendors.
- Employees has to create package with different extensions because each region regulatory standardization is different.

Initial Development

Meetings with Stakeholders:

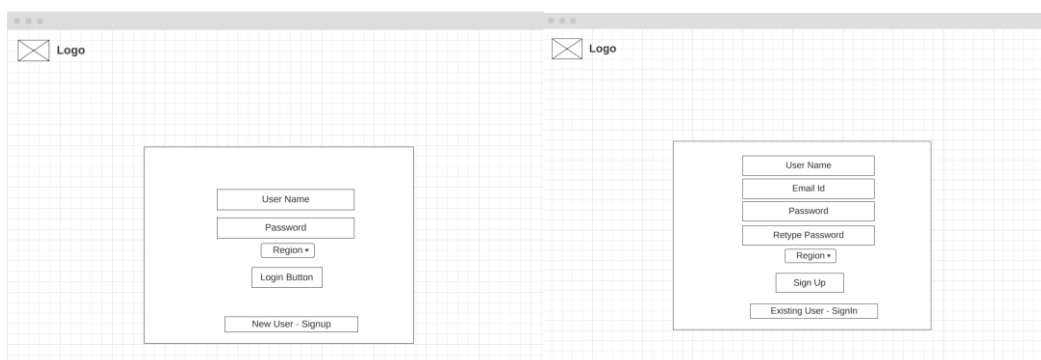
We conducted various meetings with the clients so as to bring in new features and create an initial prototype. Below were the points finalized.

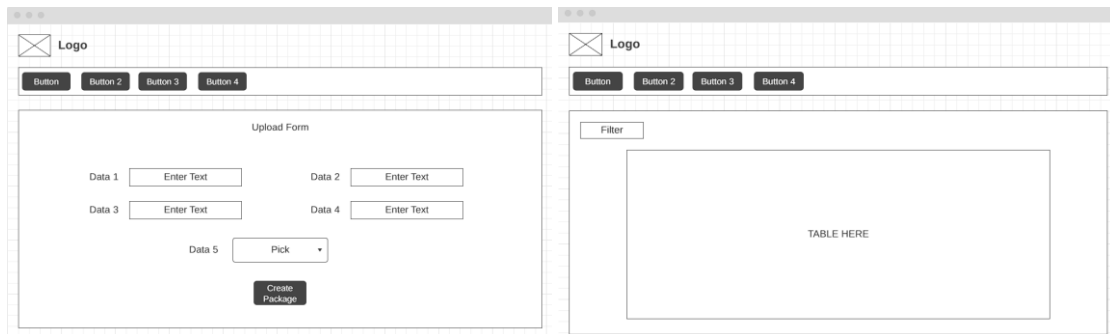
1. We learned that each region would have different set of folder/directory structure.
2. The uploading page would need a text form which can be used label the package and provide the instruction. The naming convention of the package will be based on the text fields.
3. Clients need the feature of manually entering the folder names in the text fields and automatically populating the browse path.
4. Clients required a UI which is regulatory dependant (i.e. the UI elements for packaging/uploading should be region specific).
For example: If region A has set of fields like a, b, c, d, e but region B has set of fields like b, d, e, g then the UI should be smart enough to display only required fields of A to the user who uploads the package for region A.

Based on the above meeting notes, we created an initial requirement plan:

- Tool will have 4 pages:
 - Login Page (Sign in/Sign up) – This should choose the region
 - Home Page with CTA to other pages. This will be the page to upload the files and contains CTA buttons to other pages.
 - Dashboard – This will contain data in a tabular form to display the size, date, status of each package as well as other details like user who uploaded.
 - User Preference Page – To edit the profile, so that once a region is set the data will be saved for the next use.
- Home Page will be based on the region selected. As we learned that each user will be assigned to a single region, we can set the user region as default and every time when the user log in, the data can be retrieved from Database and the default region home page will be shown, rather than selecting the region every single time. So, we have to develop Home page elements for each region. A single page has to be developed with multiple elements, which has the hidden flag to set On or Off depending on the criteria.
- There should be Page Redirect buttons as default on every page, so users can redirect to any page at any time.
- There should not be any high-resolution Image elements, as we need the UI to be stable and fast and is to be used only for the desired set of employees.
- The Dashboard to monitor the status can be filtered for other regions and also using the date range. This should be a tabular form and users can click on each data to find more details of the package.

Initial Wireframes:





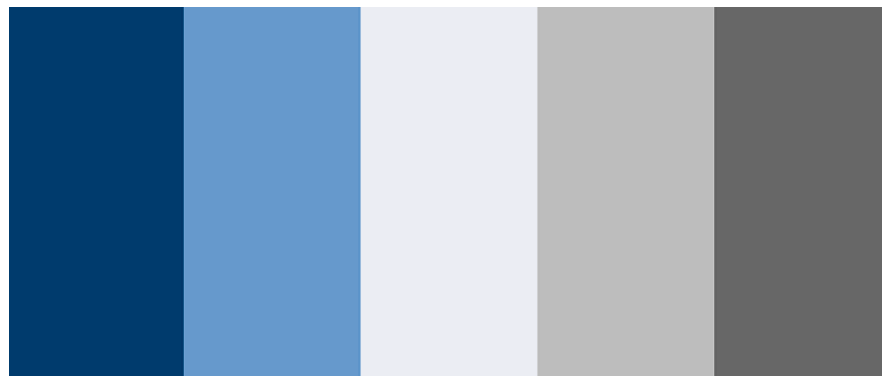
Design Theory:

As the clients did not have any existing design theory, we analysed the existing company website and noted down the Existing font elements, button styles and colour palette.

Colour Scheme Selection

We chose the Blue-Grey Colour Palette

This colour creates a peaceful atmosphere that can make website feel trustier and inviting. The grey undertones in the colour help to balance out the coolness of the blue, creating a perfect balance.



Dark Blue - #003B6D

Light Grey - #EBEDF3

Grey - #BDBDBD

Dark Grey - #676767

Typography: We implemented constant typography principle, as well as keeping the readability in mind and always contrast the button and the text

BOOK FROM

With the initial wireframes, features and the colour palette, we presented the initial demo to the clients and stakeholders. The colour palette was agreed and initial design was approved.

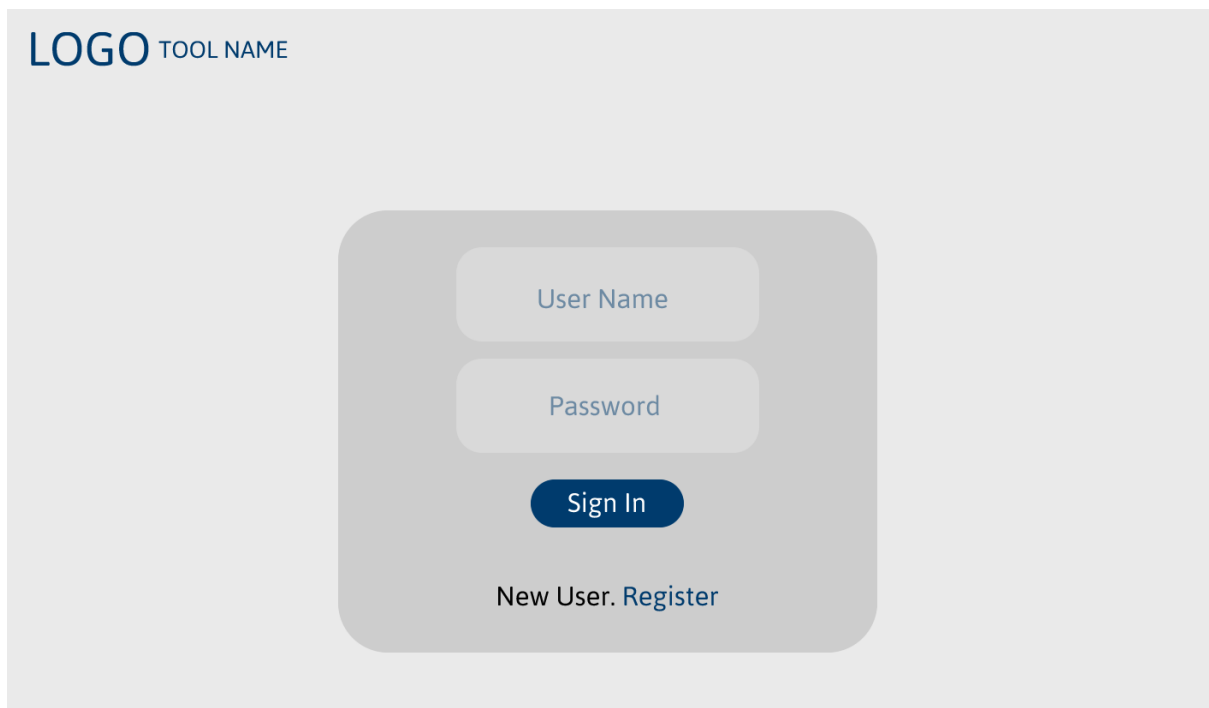
Based on our meeting Clients asked for improvisation on the below feature:

- Clients need the feature of manually entering the folder names in the text fields and automatically populating the browse path and the vice versa i.e. manually browse the required folder and text fields are populated based on the path.

High Fidelity Wireframe:



Prototype



LOGO TOOL NAME

User Name

Password

Retype Password

Region 

Register

Existing User. [Sign In](#)

LOGO TOOL NAME

USERNAME 

[Home](#) [E-Transmission](#) [Dashboard](#)

Upload Form

Data 1



Data 2

Data 3

Data 4



Data 5

Data 6

URL

Browse

UPLOAD

Upload Form

Packaging Details

Details here

Confirm

Cancel

UPLOAD



Upload Form

Packaging Details

75%

Status : Packaging in Progress

Back To Home

UPLOAD



Search

Advanced Search

TABULAR DATA

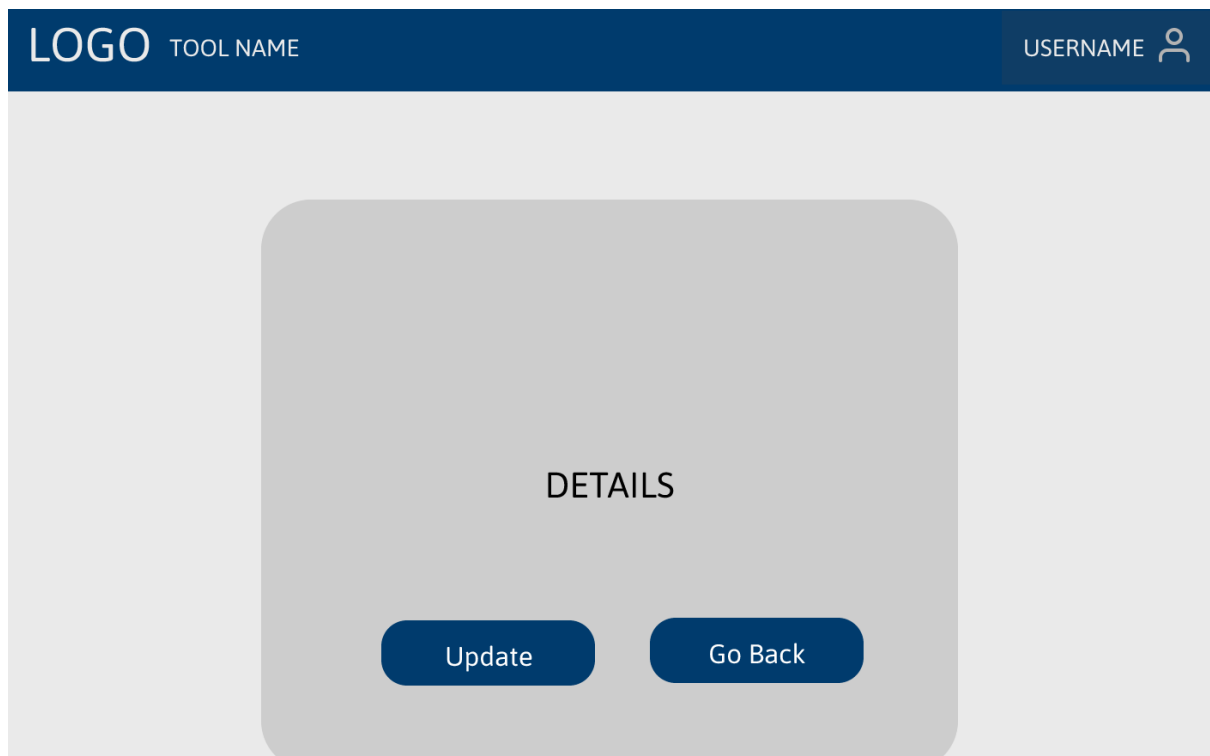
My Profile

Search

Advanced Search

Sign Out

TABULAR DATA



Once the wireframe and prototype is shown as demo to client and got approval, we started developing the UI pages

Development & Testing

The UI pages was developed using HTML/CSS/Bootstrap and JS

The upload form was made dynamic, that is depending on the Region, the input fields changed.

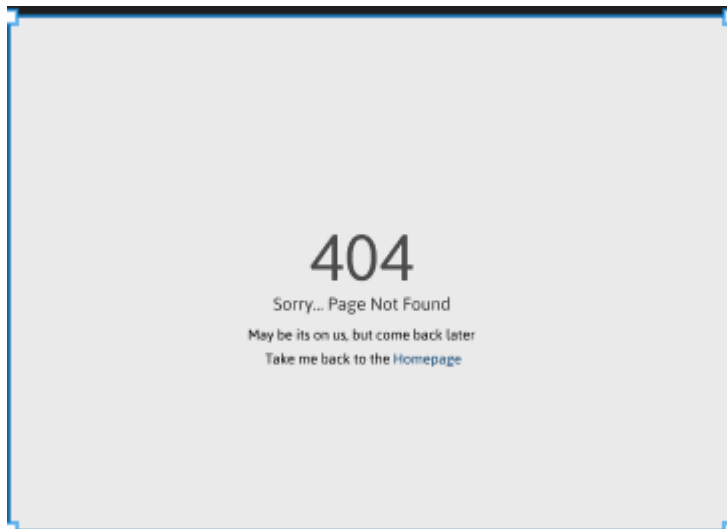
Upload Form

Data 1 Data 2 Data 3 Data 4 Data 5 Data 6 URL [Browse](#)[UPLOAD](#)

Upload Form - Region 2

Data 1 Data 2 Data 3 Data 4 URL [Browse](#)[UPLOAD](#)

All the landing pages were made along with a custom 404 error page. These pages were then integrated to the backend Spring Boot code.



Once the product was ready, we performed various unit testing and end to end User testing. The product was deployed on test and UAT environment and post the UAT sign off the product was released onto the prod env. Along with the release, we documented the Install Notes, User Manual and Design system. We also gave the initial transition to the support team on how to work on the tool.

Timeline

Requirements Phase – 1 Month

Design Phase – 1 Month

Development Phase – 2 Months

Testing Phase - .5 Month

Release - .5 Month